Where Should the Defibrillators Go?

Survival-Based, Opening-Hours-Aware and Fairness-Constrained AED Placement from Open Data in Virginia Beach

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Abstract

Out-of-hospital cardiac arrest (OHCA) claims approximately 90% of the roughly 137,000 patients reported to the U.S. CARES registry each year, with survival chances diminishing precipitously for every minute defibrillation is delayed. Public automated external defibrillators (AEDs) can bridge this critical interval, but only if devices are both spatially proximal to cardiac arrests and physically accessible when emergencies occur. Using an open municipal dataset of 2,706 cardiac arrests in Virginia Beach, Virginia (2017–2019), a 52,708-node pedestrian walking network, 929 candidate host facilities with OpenStreetMap operating schedules, and census-tract Social Vulnerability Index data, we formulate public AED deployment as an expected-survival facility location problem. We prove that the objective function is monotone submodular, guaranteeing at least $(1-1/e) \approx 63.2\%$ optimality for greedy selection ($\geq 99.9\%$ in practice), and solve it exactly via mixed-integer linear programming. In five-fold temporal cross-validation with address-clustered bootstrap intervals, our hours-aware expected-survival model achieves 42–67% greater out-of-sample survival gains than the standard maximal covering location problem across budgets of 25–200 AEDs, outperforming the baseline across all 30 sensitivity configurations and, for budgets of 50–100 AEDs, significantly outperforming stronger literature baselines. We find that assuming continuous 24/7 facility access inflates projected clinical gains by 67–106%, whereas deploying AEDs in 24/7 outdoor cabinets increases realized benefit by 41–63% for budgets of 25–200 AEDs, establishing an operational equivalence of one outdoor cabinet to approximately two indoor units. Furthermore, a spatial kernel smoother shows that geographic location explains only 7% of the variance in ambulance response times, rejecting the hypothesis that AEDs should be targeted to historically delayed EMS zones. This finding motivates an empirical response-time distribution model and yields a derived pedestrian "reach curve" showing that marginal AED utility halves within roughly 123 m of walking at a brisk pace. Finally, incorporating social vulnerability equity floors incurs negligible out-of-sample efficiency loss, while an asymptotic feasibility bound proves that spatial deficits in host commercial infrastructure preclude full parity across vulnerability quartiles using building-hosted units alone.

1 Background & Problem Definition

Clinical urgency and defibrillation delays. In the United States, emergency responders treated roughly 137,000 non-traumatic out-of-hospital cardiac arrests reported to the CARES registry in 2024 (a registry covering about 186 million people), but only 10.5% of those patients survived to leave the hospital alive [1]. For patients in ventricular fibrillation or pulseless ventricular tachycardia, survival hinges on delivering an electric countershock before irreversible ischemic injury takes hold. Every minute of delay causes a steep survival drop: in King County, Washington,

predicted survival fell by 2.3 percentage points for every minute until CPR began, and by another 1.1 points for each minute of delay to defibrillation [3].

Bystander intervention and physical access barriers. Automated external defibrillators (AEDs) allow lay bystanders to deliver early shocks guided by voice prompts. When a bystander applies an AED before EMS arrives, survival more than doubles: 24% versus 9% for bystander CPR without an AED in the Resuscitation Outcomes Consortium [7], and 66.5% versus 43.0% for observed public shockable arrests (adjusted odds ratio 2.62) [8]. Yet bystanders attach an AED in only 12.6% of public arrests nationwide [1]. In practice, bystanders face two basic physical barriers: the nearest unit is too far to walk to, or it sits behind locked doors in a building closed for the evening or weekend [10, 11, 12].

Study setting: Virginia Beach. Virginia Beach offers an unusual opportunity to study these spatial-temporal dynamics. Spanning roughly 460,000 residents across high-density coastal corridors, sprawling suburbs, and rural agricultural land, it is one of very few U.S. jurisdictions that publish complete, incident-level cardiac arrest records with the precise EMS response time for each dispatch [32]. Across these records, the median time from the 911 call to the first unit's arrival is 7.2 minutes, and 41% of patients waited longer than 8 minutes (Figure 1)—an interval where an accessible public AED could mean the difference between life and death.

Research questions. Given a municipal budget of p AEDs and a set of candidate host facilities, we examine four core questions:

RQ1 Does optimizing site selection for *expected survival* over actual walking networks yield greater benefit on held-out arrests than standard models that maximize historical arrests covered within a fixed 100-meter buffer?

RQ2 How severely does the assumption of continuous 24/7 building access inflate projected survival gains, and what share of that lost benefit can outdoor 24/7 cabinets recover?

RQ3 Can historical ambulance delays guide prospective placement, or is dispatch latency too spatially variable to predict where AEDs will be needed next?

RQ4 What is the price of fairness when requiring every social vulnerability quartile to receive an equitable share of expected survival, and does candidate site density impose a hard mathematical ceiling on parity?

2 Literature Review

Survival and time. The link between time and survival is among the best-established facts in resuscitation science. Larsen et al. fit a linear model to 1,667 witnessed ventricular-fibrillation arrests and found that predicted survival falls from 67% by 2.3, 1.1 and 2.1 percentage points for every minute of delay to CPR, defibrillation and advanced life support, respectively [3]. Valenzuela et al. replaced this with a logistic model that depends only on the collapse-to-CPR and collapse-to-shock intervals and generalizes across U.S. cities [4]. De Maio et al. fit a similar curve to EMS response time for all treated arrests in Ontario [5]. Erkut, Ingolfsson and Erdoğan showed that replacing yes/no "coverage" with such survival functions changes, and improves, where ambulances should be based [6]; we use their published forms of both curves.

Bystander AEDs work, but are rarely used. Weisfeldt et al. found that an AED was applied before EMS arrival in only 2.1% of 13,769 arrests, but that its use raised survival (adjusted OR

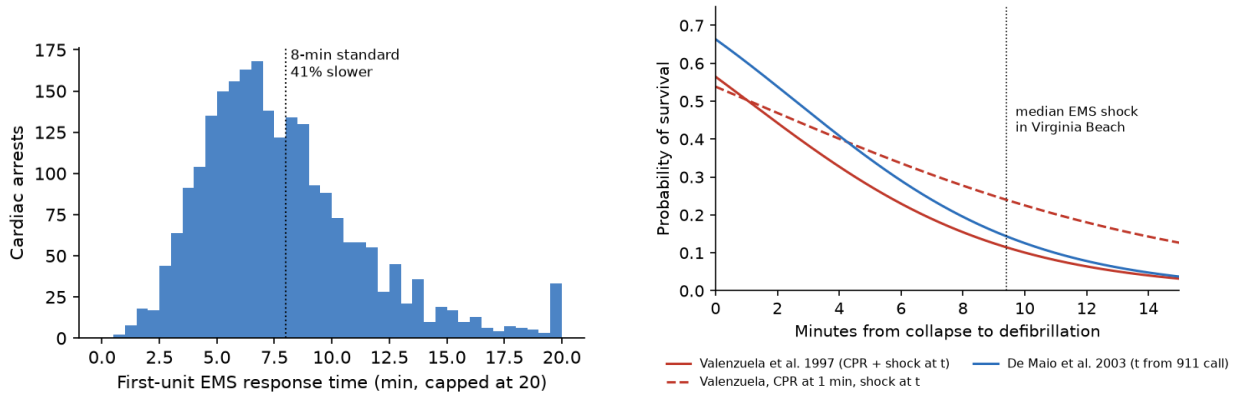


Figure 1: Left: first-unit EMS response times for the 2,332 Virginia Beach arrests with a recorded time. **Right:** the two published survival curves used in this study [4, 5, 6]. A shock at the median Virginia Beach EMS time (dotted line) gives far lower survival than a shock in the first few minutes. Valenzuela's curve is measured from collapse; De Maio's from the 911 call.

1.75) [7]. Pollack et al. found that the benefit of a bystander shock "increased progressively as EMS response time became longer" [8], and Hindborg et al. found a significant survival benefit specifically when EMS took 7–9 minutes in public arrests [9]. These findings motivate RQ3.

Access in space and time. Copenhagen data showed that 61.8% of public arrests happen in the evening, at night or on weekends, while only 9.1% of AEDs were accessible 24/7; limited hours cut coverage by 53.4% during those periods [10]. When the nearest AED was accessible, 30-day survival was 28.8% versus 16.4% when it was not [11]. In England, only 34.3% of AEDs were available at night [12]. Walking routes matter too: switching from straight-line to route distance cut 100-metre coverage from 24.2% to 9.5% in Copenhagen [13].

Optimization models. Toregas et al. introduced the location set-covering problem [14] and Church and ReVelle the *maximal covering location problem* (MCLP): place p facilities to cover as much demand as possible within a fixed distance [15]. A 2025 scoping review of 43 AED-placement studies found the MCLP to be the most common model [17]. Chan et al. first applied optimization to AED siting in Toronto, using historical arrests and a 100-metre radius [18, 19]. Sun et al. added time-of-day availability and found that assuming 24/7 access overstated coverage by 21.5% [20]; an *in silico* trial in Copenhagen found optimized placement covered 24.2% of arrests versus 15.8% for guideline-based placement [21]. Siddiq et al. studied how the coverage radius changes results [22]. Pourghaderi et al. replaced coverage with a survival-maximizing objective in Singapore [23], and Buter et al. used distance-decaying "gradual" coverage with greedy heuristics [24]. Other studies examined outdoor and landmark-based deployment in Paris [25], hotspot relocation in Switzerland [26], and cost-effectiveness in England [27].

Equity. Bystander CPR is far less likely in low-income Black neighborhoods (OR 0.49) [28], and both lay AED use and survival fall as the share of Black residents rises [29]. In England, arrests are concentrated in deprived areas while AEDs are concentrated in affluent ones [27]. Leung et al. showed that optimizing for coverage already shifted AEDs toward deprived Scottish neighborhoods [30], but no AED study we found imposes an explicit fairness constraint or reports its cost. Bertsimas, Farias and Trichakis define the *price of fairness*: the fraction of total benefit given up to

achieve a fair allocation [31].

Gap. To our knowledge, no prior study (i) tests whether an arrest's own observed EMS response time should weight AED demand, (ii) imposes and prices a fairness constraint by CDC Social Vulnerability Index (SVI) quartile [33], (iii) derives the distance-decay of an AED's value from a city's own EMS response-time distribution instead of assuming it, or (iv) combines survival-based scoring, real walking routes and opening hours on a public U.S. city data set with temporal cross-validation against several published baselines.

3 Proposed Solution / Theoretical Framework

3.1 From coverage to expected survival

Let I be the set of historical arrests and J the candidate host sites. For arrest i let r_i be the EMS response time (minutes), t_i the time of the 911 call, and d_{ij} the walking distance from the arrest to site j. Let $a_j(t) \in \{0, 1\}$ indicate whether site j is open at time t.

Timeline. Measured in minutes from collapse, the 911 call is placed after a delay c. Without an AED, CPR begins when EMS arrives, at $c + r_i$, and the first shock follows δ minutes later. With an AED at distance d_{ij}, a bystander walks there and back at speed v and needs σ minutes to apply the pads, so the AED is ready at

$$\tau_{ij} = c + \frac{2d_{ij}}{60\,v} + \sigma.$$

Whichever comes first, AED or EMS, starts the treatment.

Survival gain. Using Valenzuela's logistic model [4], as reproduced by [6],

$$S(T_{\text{CPR}}, T_{\text{shock}}) = \left(1 + e^{-0.260 + 0.106\,T_{\text{CPR}} + 0.139\,T_{\text{shock}}}\right)^{-1},$$

the gain in survival probability if arrest i is served by site j is

$$g_{ij} = a_j(t_i)\Big[S\big(\min(\tau_{ij}, c + r_i),\ \min(\tau_{ij}, c + r_i + \delta)\big) - S(c + r_i,\ c + r_i + \delta)\Big]$$

when $d_{ij} \le D_{\max}$, and 0 otherwise. Thus $g_{ij} = 0$ automatically if the site is closed, if it is too far, or if EMS would arrive first. The model assumes CPR starts when the AED or EMS arrives, whichever is first. Default parameters are listed in Table 1.

Response time as a random variable. When planning prospective AED placements, the response latency r_i for a future incident is naturally unknown. While one might initially weight demand by each historical arrest's observed response time, Section 5.3 demonstrates that this provides no generalization benefit, because location predicts ambulance delay very poorly. Our primary formulation therefore models the response time as a random variable R sampled from the municipality's empirical response-time distribution (fitted strictly on training folds), scoring each arrest–site pair by its expected marginal survival gain:

$$\bar{g}_{ij} = a_j(t_i)\,\rho(d_{ij}), \qquad \rho(d) = \mathbb{E}_R\big[\,g(d, R)\,\big],$$

where $g(d, r)$ denotes the bracketed survival gain defined above. The resulting scalar function $\rho(d)$, which we term the *reach curve*, quantifies the expected clinical utility of an AED as a direct function

of walking distance alone. Rather than postulating an ad-hoc distance-decay parameter, the reach curve is derived directly by integrating clinical survival dynamics over empirical dispatch delay distributions. Because aggregate response-time distributions are routinely published by municipal EMS agencies, this formulation transfers readily to jurisdictions without incident-level microdata. Because $\bar{g}_{ij}$ depends only on distance and opening hours, the main model is a gradual-coverage MCLP [19, 24]; what is new is that its distance decay is derived from a survival curve and the city's own EMS delays rather than chosen by hand.

Symbol	Default	Meaning and source
v	1.4 m/s	brisk walking speed of the bystander (tested 1.0 and 2.0)
σ	0.5 min	time to open the cabinet, power on and attach pads (tested 1.5)
c	1 min	collapse to 911 call (tested 2 min)
δ	1 min	EMS arrival to first shock (tested 2 min)
$D_{\max}$	500 m	one-way walking limit for a retrieval (tested 300 and 800 m)
$S(\cdot)$	Valenzuela	survival curve [4, 6] (tested De Maio [5], whose predictor is the interval from the 911 call)

Table 1: Model parameters. Every default is varied in the sensitivity analysis (Section 5.5).

3.2 The optimization problem

Each arrest uses the best open AED nearby, so the expected survival gain of a set of sites $X \subseteq J$ is

$$F(X) = \sum_{i \in I} \max_{j \in X} g_{ij} \qquad (\text{with } \max_{\emptyset} = 0),$$

and we solve $\max\{F(X) : |X| \le p\}$. As a mixed-integer linear program, with $y_j = 1$ if site j gets an AED and x_{ij} the share of arrest i served by site j:

$$\max \sum_{i,j} g_{ij} x_{ij} \quad \text{s.t.} \quad \sum_j y_j \le p, \quad x_{ij} \le y_j, \quad \sum_j x_{ij} \le 1, \quad y_j \in \{0,1\},\ x_{ij} \in [0,1].$$

For fixed y, the best x puts all of arrest i on its best open site, so the program computes exactly $\max F$. The classic MCLP is the special case $g_{ij} = 1$ if $d_{ij} \le 100\,\text{m}$ and 0 otherwise, with every site assumed always open.

3.3 A guarantee for the greedy algorithm

Definition 1. *A set function F is* monotone *if $F(A) \le F(B)$ whenever $A \subseteq B$, and* submodular *if adding a site helps a small set at least as much as a large one: $F(A \cup \{k\}) - F(A) \ \ge \ F(B \cup \{k\}) - F(B)$ for all $A \subseteq B$ and $k \notin B$.*

Theorem 1. *F is monotone and submodular with $F(\emptyset) = 0$. Hence the greedy algorithm, which repeatedly adds the site with the largest marginal gain, returns a set X_G with $F(X_G) \ge \big(1 - (1 - \frac{1}{p})^p\big)\,\text{OPT} \ge (1 - \frac{1}{e})\,\text{OPT} \approx 0.632\,\text{OPT}$.*

Proof. Write $F = \sum_i f_i$ with $f_i(X) = \max_{j \in X} g_{ij}$, where all $g_{ij} \geq 0$. Monotonicity: enlarging X can only raise a maximum. Submodularity: fix $A \subseteq B$ and $k \notin B$, and let $m_A = f_i(A) \leq m_B = f_i(B)$. Adding k changes f_i by $\max(0, g_{ik} - m_A)$ for A and by $\max(0, g_{ik} - m_B)$ for B. Since $m_A \leq m_B$, the first is at least the second. A sum of monotone submodular functions is monotone submodular, so F is too. The approximation ratio is the theorem of Nemhauser, Wolsey and Fisher [16]. The proof only uses $g_{ij} \geq 0$, so it covers every variant in this paper: opening hours (which set $g_{ij} = 0$ when a site is closed) and the expected gains $\bar{g}_{ij}$ (which are non-negative because each $g(d, r)$ is). □

In practice, greedy reached at least 99.9% of the exact optimum in all ten instances we checked (indoor and cabinet models, $p = 10$ to 200), so the bound is very conservative here. We use greedy's ordered site list in the interactive prototype (Section 7), since it naturally answers "which AED should the city buy next?".

3.4 Fairness and a feasibility bound

Let $Q_1, \ldots, Q_4$ be the arrests in each SVI quartile of census tracts (Q_4 most vulnerable). For the unconstrained optimum $X^\star$ with p AEDs, let T_p be its citywide average gain per arrest (over the arrests that have an SVI score). We write T when p is clear. A plan is *φ-fair* if every quartile receives at least a fraction φ of that average:

$$\frac{1}{|Q_q|} \sum_{i \in Q_q} \max_{j \in X} g_{ij} \;\geq\; \varphi T \qquad \text{for } q = 1, \ldots, 4.$$

These are linear constraints in the MILP. The *price of fairness* is $1 - F(X_\varphi)/F(X^\star)$ [31]. We use a fixed target T rather than the plan's own average: otherwise, the solver could appear fairer by "declining" to use benefit in well-served areas, which is not how real AEDs work.

Proposition 1. *Let $C_q = \frac{1}{|Q_q|} \sum_{i \in Q_q} \max_{j \in J} g_{ij}$ be the most any plan could give quartile q, with an AED at* every *candidate site. A φ-fair plan with p AEDs can exist only if $\varphi \leq \min_q C_q / T_p$.*

Proof. For any $X \subseteq J$, $\max_{j \in X} g_{ij} \leq \max_{j \in J} g_{ij}$, so quartile q receives at most C_q. Fairness requires this to be at least φT_p. □

This condition provides an upper bound that is necessary rather than sufficient: at modest budgets, fairness can also fail simply because p is too small to distribute coverage broadly across the city. The budget-independent asymptotic form provides the clearest policy takeaway. Let T_∞ denote the citywide mean survival gain per arrest when an AED is placed at every candidate host facility in the city. Then no individual quartile q can ever achieve more than a fraction C_q/T_∞ of that theoretical maximum, regardless of budget size. Where a neighborhood exhibits a structurally sparse built environment with few candidate properties, mathematical site selection alone cannot overcome this geographic bottleneck, demonstrating that municipal equity requires complementary interventions such as 24/7 outdoor street-side cabinets, dispatched volunteer responders, or drone delivery.

4 Experimental Design & Implementation

4.1 Data

- **Cardiac arrests:** all 2,706 OHCA dispatches in Virginia Beach from 1 January 2017 to 30 June 2019, with call time, location (rounded to the block) and first-unit response time [32]. 373

records list a response time of exactly 0 and one lists 128 minutes; we treat these as missing and impute them from a linear fit on road distance to the nearest of the city's 40 first-responder stations (fit on the 2,332 valid records; $r = 0.21$). The main model's response-time distribution uses only valid records, and we repeat the main comparison with the imputed records removed.

- **Candidate sites:** 929 public places from OpenStreetMap (Overpass API): places of worship (218), convenience stores (134), fuel stations (128), schools (120), banks (72), supermarkets (70), fitness and sports centres (77), fire and police stations (38), pharmacies (27), post offices (25), libraries (12) and others.
- **Opening hours:** 209 sites have machine-readable OpenStreetMap hours; for the rest we assume typical hours by category (e.g. schools weekdays 7:00–16:00; places of worship Sunday morning and Wednesday evening; fire and police stations 24/7).
- **Walking distances:** shortest paths on the OpenStreetMap pedestrian network of Virginia Beach (52,708 intersections, 71,677 street segments) via OSMnx, up to 1 km. Each distance includes the walk from the arrest to the network and from the network to the site, and is never shorter than the straight line. The 51 arrests and 46 sites more than 150 m from any walkable path are treated as unreachable on foot. Only 3.1% of arrests are within 100 m of *any* candidate site, 21.0% within 300 m and 40.5% within 500 m.
- **Vulnerability:** CDC/ATSDR Social Vulnerability Index 2022 [33] for the city's 112 census tracts (2022 TIGER boundaries; 111 have an SVI score), split into quartiles within the city.

4.2 Experimental procedure

1. **Benchmarked methods.** We evaluate seven deployment strategies: (a) random site selection averaged over 30 uniform draws; (b) the classic MCLP [15], which maximizes arrests covered within 100 m of walking under the assumption of unconstrained 24/7 access; (c) hours-constrained MCLP, which incorporates opening schedules into the 100 m covering radius following Sun et al. [20]; (d) extended-radius MCLP using a 300 m walking threshold; (e) unconstrained survival optimization without opening hours, following Pourghaderi et al. [23]; (f) our proposed expected-survival model integrating pedestrian routing and opening schedules; and (g) our model assuming all deployed devices are housed in outdoor 24/7 cabinets. Each strategy selects exactly p locations; whenever an algorithm exhausts sites offering non-zero marginal coverage, remaining devices are allocated to candidate facilities with the highest historical arrest incidence.
2. **Temporal cross-validation.** The 30-month observational window is divided into five sequential six-month folds. To evaluate out-of-sample generalizability, models are trained on four folds and evaluated on the single held-out fold, accumulating out-of-sample performance across all 2,706 incidents. The final fold (training on 2017–2018, testing on the first half of 2019) provides a prospective, forward-in-time evaluation reflecting realistic municipal deployment, and is analyzed independently.
3. **Evaluation metrics.** All candidate policies are evaluated with a common model-based measure: realized expected survival gain summed across out-of-sample incidents, conditioned on each event's actual EMS response time, 911 call time, pedestrian network distance, and the host site's opening hours (recorded or assumed by category). To ensure a balanced assessment against models that do not optimize clinical survival directly, we also compute the classic covering metric: the proportion of out-of-sample arrests situated within 100 m walking distance of an accessible AED.
4. **Statistical uncertainty.** Because approximately 70% of cardiac arrests in the dataset recur at shared street addresses, standard independent and identically distributed bootstrap resam-

pling artificially narrows error margins. We therefore report 95% confidence intervals computed from 2,000 cluster-bootstrap iterations grouped by unique street address, evaluating pairwise performance differentials across paired resamples.

5. **Optimization implementation.** Mixed-integer linear programs are solved to exact optimality using HiGHS via SciPy, complemented by lazy greedy heuristic evaluations. Because multiple site configurations may yield identical objective values, a small secondary tie-breaker prioritizes sites exhibiting higher aggregate arrest density within the training split; this tie-breaker operates exclusively on training folds and never accesses held-out incidents.

4.3 Units: "potential survivors"

Because Valenzuela's curve describes witnessed shockable arrests, F counts *potential survivors*: extra survivors if every arrest were witnessed, shockable, and had a willing bystander. Nationally, 10.8% of arrests are bystander-witnessed and shockable [2], so multiplying by 0.108 gives a rough real-world estimate. Even this is an upper bound, because not every bystander will fetch an AED and many arrests happen in homes. We report potential survivors because they compare methods fairly; absolute lives are discussed in Section 6.

5 Results

5.1 RQ1: survival-based placement generalizes better

Method	$p = 10$	25	50	100	200
Random sites	0.8	1.5	3.1	6.1	11.8
Classic MCLP (100 m, assumes 24/7) [15]	3.2	8.3	11.4	14.7	19.5
MCLP 100 m + opening hours (cf. [20])	4.8	8.9	11.2	14.8	19.9
MCLP 300 m	5.5	9.2	11.6	17.4	24.4
Survival, no opening hours (cf. [23])	5.7	11.2	15.1	22.8	27.1
Expected survival + hours (ours)	5.3	**11.7**	**18.6**	**24.5**	**28.4**
Ours − classic	+2.1	+3.5	+7.2	+9.8	+8.9
95% CI (address-clustered)	[−0.6, 4.8]	[0.2, 7.3]	[2.8, 11.9]	[5.3, 14.8]	[4.6, 13.8]

Table 2: Out-of-sample expected survival gain (potential survivors over 30 months), pooled across five temporal folds. Every method places exactly p AEDs.

Our expected-survival framework consistently outperforms the classic MCLP across all evaluated installation budgets (Table 2, Figure 2). Relative to standard covering placement, the proposed model increases out-of-sample survival gains by 42% at $p = 25$, 63% at $p = 50$, 67% at $p = 100$, and 46% at $p = 200$. These advantages remain statistically significant for all budgets $p \geq 25$ under clustered bootstrap resampling on street addresses. The model also surpasses stronger formulations from the operational literature: at moderate and large budgets ($p = 50$ and $p = 100$), it significantly exceeds the performance of hours-aware covering models, 300-meter extended covering formulations, and unconstrained survival scoring that neglects operating schedules. At minimal deployment budgets ($p = 10$), performance differences across non-random methods remain within statistical margins of error.

These clinical advantages hold under prospective forward validation. When trained strictly on 2017–2018 data and evaluated prospectively on the first half of 2019, our model exceeds classic MCLP performance across all five budgets (1.8 vs. 1.2 at $p = 10$, 3.5 vs. 2.1 at $p = 25$, 4.5 vs. 2.2

at $p = 50$, 6.7 vs. 2.3 at $p = 100$, and 7.0 vs. 3.0 at $p = 200$), prevailing in 23 of the 25 individual fold-by-budget pairings. On simple 100 m coverage the picture is mixed. Counting only AEDs open at the time of the arrest, our plan covered as many or more held-out arrests for $p \geq 50$ (26 vs. 25 at $p = 50$; 32 vs. 28 at $p = 100$). Ignoring opening hours, which is the classic method's own measure, the classic plan covered more at small budgets (20 vs. 7 at $p = 10$, 34 vs. 20 at $p = 25$, 43 vs. 40 at $p = 50$), and the two were tied from $p = 100$ onward (48 vs. 47; 53 vs. 53). The structural limitation of the classic MCLP stems from pedestrian network sparsity: only 3.1% of cardiac arrest locations fall within 100 m walking distance of any candidate host facility. Enforcing a strict 100-meter threshold discards incidents located at moderate walking distances (100–300 m) where rapid bystander retrieval could still successfully precede emergency medical arrivals.

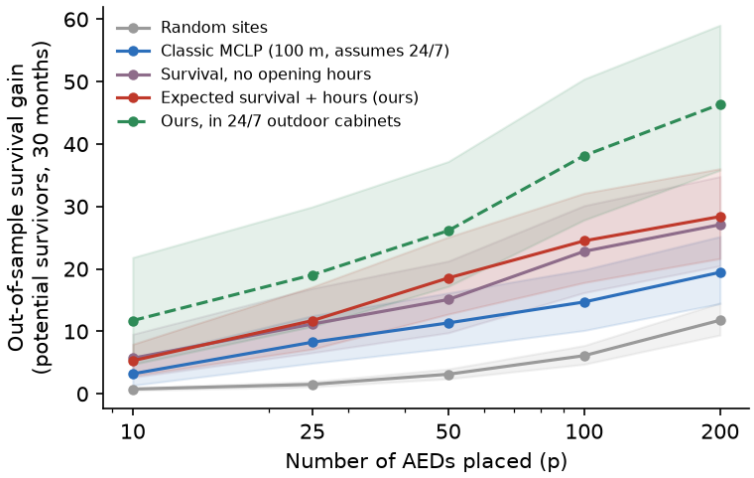


Figure 2: Out-of-sample expected survival gain with 95% address-clustered bootstrap bands. The dashed line is our model with every AED in a 24/7 outdoor cabinet.

5.2 RQ2: opening hours matter more than expected

p	Promised by hours-blind plan	Delivered by that plan	Hours-aware (ours)	Ours in cabinets
25	19.1	11.2	11.7	19.1
50	26.1	15.1	18.6	26.1
100	38.2	22.8	24.5	38.2
200	46.4	27.1	28.4	46.4

Table 3: Out-of-sample expected survival gain (potential survivors) when planners ignore or use opening hours. Without opening hours, our model is the same as the cabinet model, so the hours-blind plan's promise is exactly what it would deliver if its AEDs were in 24/7 cabinets.

Assuming continuous 24/7 facility access introduces substantial optimism bias into municipal AED planning. Across held-out cardiac arrests, an optimization model that ignores opening hours projects 67–73% more clinical benefit than it ultimately delivers across practical budgets ($p = 25$–200), and 106% more at $p = 10$ (Table 3, Figure 4). This accessibility deficit substantially exceeds the 21.5% overestimation documented in metropolitan Toronto [20]. Explicitly embedding temporal availability schedules into site selection recovers a meaningful share of this deficit, providing a 23% increase in realized survival gain at $p = 50$ (18.6 vs. 15.1) and a 7% increase at $p = 100$ (24.5 vs.

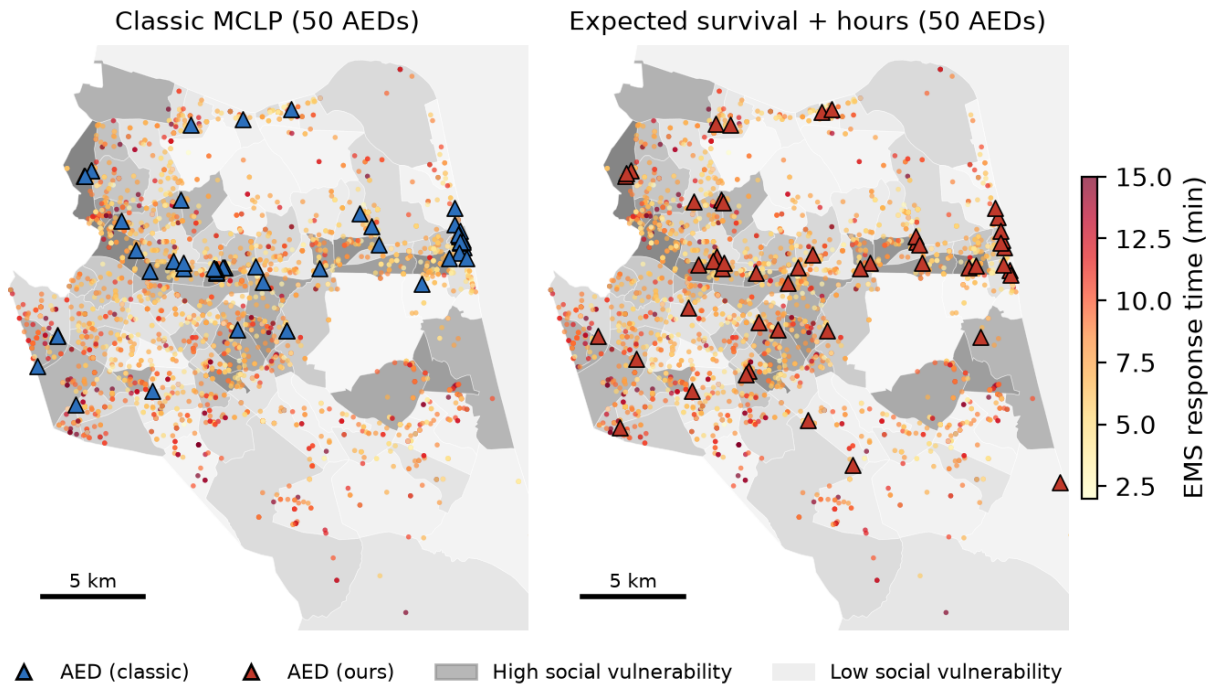


Figure 3: Fifty AED sites chosen by the classic MCLP (left) and by our model (right), trained on 2017–18. Dots are cardiac arrests colored by EMS response time; tract shading is the Social Vulnerability Index. Blank areas had no recorded arrests in the study period.

22.8), both statistically significant. However, physical infrastructure represents a far more potent lever: housing devices in weatherproof, 24/7 accessible outdoor cabinets increases out-of-sample survival gains by 41–63% over indoor placement across all budgets from $p = 25$ to 200.

The clinical penalty of indoor placement is heavily concentrated during nighttime hours: between midnight and 3:00 a.m., indoor installations capture only approximately one-fifth of the survival gain provided by outdoor cabinets, and between 3:00 a.m. and 6:00 a.m. their realized utility drops nearly to zero. As illustrated in Figure 5 (right), this operational divergence suggests a direct policy equivalence: a deployment of 25 outdoor cabinets delivers equivalent out-of-sample survival gains to 50 indoor AEDs (19.1 vs. 18.6), while 100 cabinets substantially outperform 200 indoor units (38.2 vs. 28.4). Consequently, one outdoor 24/7 cabinet provides the effective clinical utility of roughly two indoor AEDs, which gives a break-even cost ratio: a cabinet is worth choosing when it costs less than about twice an indoor installation, before accounting for maintenance and access arrangements.

5.3 RQ3: steering AEDs toward slow-EMS areas does not help

Our initial formulation explored weighting candidate locations by historical ambulance delays, incorporating incident-specific response times into each demand node. We conducted an ablation study (Table 4) to assess whether this spatial delay signal enhances generalization, comparing incident-specific response times against a uniform citywide median latency and our empirical distribution formulation.

Conditioning demand on historical incident-level response times does not improve out-of-sample generalizability; across all evaluated budgets, performance differences among the three variants are

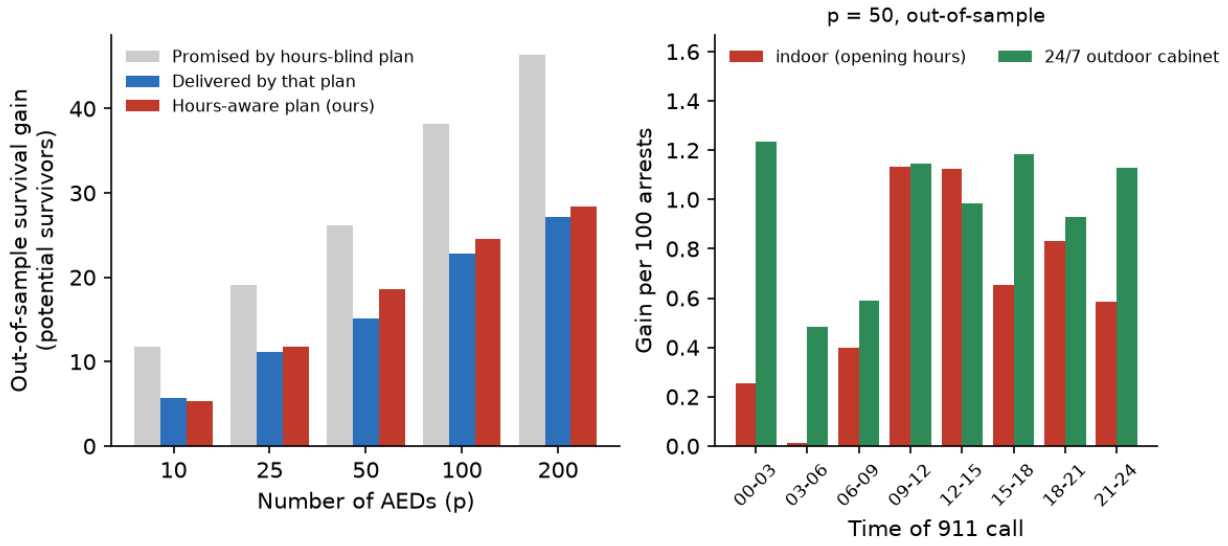


Figure 4: Left: out-of-sample benefit an hours-blind AED plan promises versus what it delivers, and what an hours-aware plan delivers. **Right:** out-of-sample benefit by time of the 911 call for 50 AEDs, indoors versus in 24/7 outdoor cabinets. The indoor and cabinet plans are optimized separately and use different sites, which is why indoor can exceed cabinets in some time blocks.

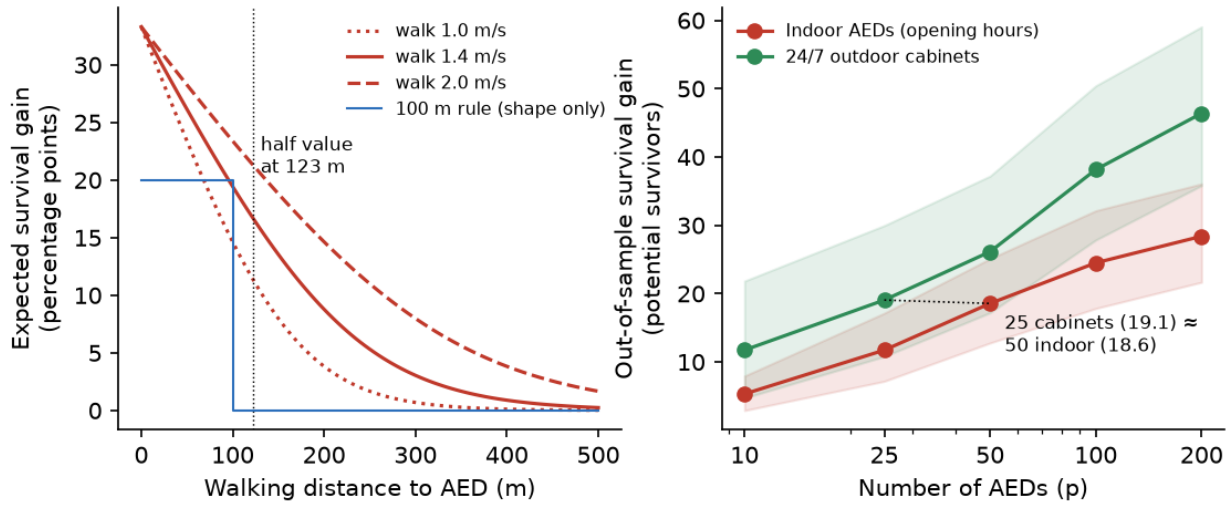


Figure 5: Left: the reach curve $\rho(d)$, the expected survival gain from an AED at walking distance d, averaged over Virginia Beach's real EMS response times, for three walking speeds. The blue step shows the shape of the classic 100 m rule (its height is arbitrary). **Right:** out-of-sample benefit of indoor AEDs (with opening hours) versus 24/7 outdoor cabinets, with 95% bands.

Variant (all use survival scoring and opening hours)	$p = 25$	$p = 50$	$p = 100$
Each arrest's own response time	13.3	17.3	24.0
Median response time for every arrest	12.2	19.9	24.7
Response time *distribution* (main model)	11.7	18.6	24.5

Table 4: Ablation: out-of-sample potential survivors. None of the pairwise differences is statistically significant (address-clustered 95% CIs all include zero).

statistically indistinguishable. Incorporating spatially smoothed response surfaces (such as a 1 km Gaussian kernel estimate fitted solely on training splits) yields no significant improvement. This is because ambulance response latency in Virginia Beach is largely unpredictable from spatial coordinates alone: a leave-one-out spatial kernel smoother accounts for at most 7% of response-time variance ($R^2 = 0.07$ at a 1 km bandwidth). Rather than reflecting persistent geographical bottlenecks, dispatch delays likely depend more on circumstances such as which unit is free and traffic conditions, which our data do not record. While resuscitation studies find that the benefit of bystander defibrillation is larger when ambulances are slower [8], with a significant benefit at EMS response times of 7–9 minutes in public arrests [9], historical delay patterns cannot reliably identify where future ambulance delays will occur.

This negative finding yields a valuable methodological simplification. Because ambulance latency behaves as an essentially location-invariant random variable across urban space, prospective optimization can integrate directly over the municipality's empirical response-time distribution, yielding the pedestrian reach curve $\rho(d)$ shown in Figure 5 (left). In Virginia Beach, an AED positioned immediately adjacent to a collapse provides an expected 33 percentage-point increase in survival probability for witnessed shockable arrests. However, this expected clinical utility halves within roughly 123 m of walking distance at a brisk pace (88 m at a slow 1.0 m/s), drops to 9% of its maximum value at 300 m, and approaches zero beyond 500 m. This decay profile elucidates why conventional 100-meter guidelines capture the relevant spatial scale, while highlighting the inefficiency of sharp coverage cutoffs that award identical full credit at 99 m and zero credit at 101 m. The superior generalization of our framework arises from coupling this continuous clinical distance decay with empirical operating schedules (Table 2), properly capturing viable retrievals in the 100–250 m range while ensuring that selected facilities remain accessible when emergencies occur.

5.4 RQ4: fairness is cheap, but full parity is impossible

In unconstrained survival optimization, the lowest per-arrest coverage falls upon the two least socially vulnerable quartiles rather than disadvantaged communities (Figure 6). Specifically, Q_2 experiences the lowest per-incident survival gain at budgets $p = 25$ and $p = 50$, while Q_1 is least served at $p = 100$. These quartiles correspond primarily to suburban tracts and rural southern regions characterized by low commercial density: Q_2 encompasses 474 km^2 with only 0.65 candidate host sites per km^2, while Q_1 contains 0.88 sites per km^2 and exhibits the slowest baseline ambulance response (median 7.7 minutes). In contrast, the two highest vulnerability quartiles exhibit approximately 3.0 candidate sites per km^2. The most vulnerable quartile (Q_4) attains above-average per-arrest survival gains at $p = 50$ and $p = 100$ (though not at $p = 25$). Thus, unlike findings reported in metropolitan England [27], spatial deficits in Virginia Beach track low-density suburban geography more closely than social vulnerability.

Enforcing social vulnerability equity constraints imposes negligible efficiency losses on municipal systems. On in-sample data, requiring every vulnerability quartile to attain at least 90% of the citywide mean survival gain reduces aggregate benefit by merely 0.1% at $p = 50$ and 2.6% at $p = 25$; raising the threshold to 95% parity at $p = 50$ incurs an efficiency loss of 0.9%. On out-of-sample evaluations, the cost of fairness was within noise (we did not compute a confidence interval for this comparison). At $p = 50$, establishing a fairness floor of $\varphi = 0.85$ increases the expected survival rate in the least-served quartile from 0.46 to 0.62 per 100 arrests, maintains survival in the highest-vulnerability quartile virtually unchanged (0.80 vs. 0.79), and delivers equivalent or slightly higher citywide out-of-sample benefit (19.7 vs. 18.6 potential survivors).

Nevertheless, structural urban geography establishes a strict mathematical ceiling that prevents

full parity across vulnerability quartiles. Even if an AED were installed at every candidate facility in the municipal catalog, the quartile least served at $p = 100$ (Q_1) could attain a maximum expected gain of $C_1 = 0.0123$ per arrest, compared to an all-sites citywide average of $T_\infty = 0.0153$. Applying Proposition 1, no placement policy can ever give the least-served quartile (Q_1) more than $C_1/T_\infty = 0.0123/0.0153 \approx 81\%$ of this maximal citywide benchmark, irrespective of budget allocation. For finite budgets, feasibility is bounded by C_1/T_p: at $p = 100$, this ratio equals $0.0123/0.0138 \approx 0.89$, which rules out $\varphi = 0.90$; consistent with this, the solver found $\varphi = 0.90$ infeasible and $\varphi = 0.85$ feasible. At modest budgets ($p = 25$ and $p = 50$), $C_1/T_p > 1$, indicating that equity in smaller programs is restricted by device availability rather than candidate site density.

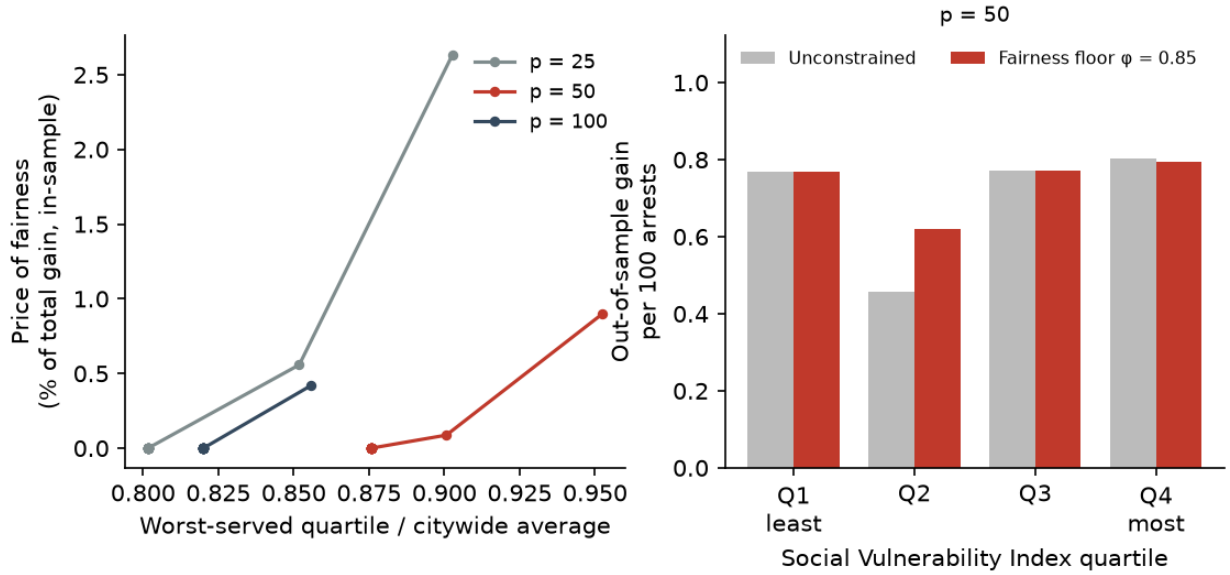


Figure 6: Left: in-sample price of fairness as the worst-served quartile is raised toward the citywide average. **Right:** out-of-sample benefit per 100 arrests by SVI quartile for 50 AEDs, without and with a fairness floor.

5.5 Robustness

To evaluate model stability under varying clinical and operational assumptions, we repeated the five-fold cross-validation across ten alternative sensitivity scenarios: modified walking speeds (1.0 m/s and 2.0 m/s), altered pedestrian distance limits (300 m and 800 m), the alternative De Maio survival function [5], an extended two-minute call delay, an extended 1.5-minute pad application latency, an extended two-minute EMS shock interval, exclusion of the 374 imputed ambulance response records, and exclusion of the 20 highest-frequency dispatch addresses (which include at least one clinic and a hospital campus). Our proposed expected-survival model out-performed the classic MCLP across all 30 budget-and-scenario evaluations ($p = 25, 50, 100$), with 95% address-clustered bootstrap intervals strictly excluding zero in 28 of the 30 configurations (Figure 7). The only two instances where confidence intervals crossed zero occurred at the smallest budget ($p = 25$): when restricting pedestrian reach to 300 m, and when excluding the top 20 repetitive dispatch addresses. De Maio's curve, as reproduced by Erkut et al. [6], gives higher absolute survival than De Maio's own population rates, so we use it only to check that the ranking of methods is unchanged.

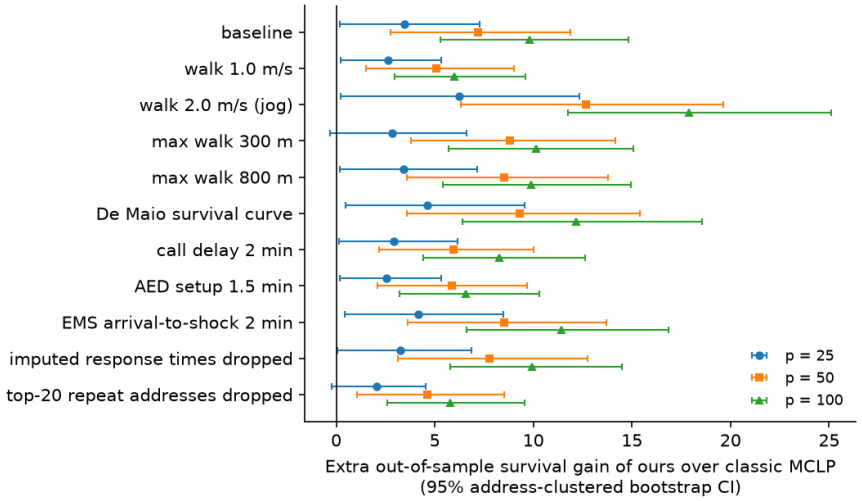


Figure 7: Out-of-sample advantage of our model over the classic MCLP under alternative assumptions (95% address-clustered bootstrap intervals).

6 Discussion

Policy guidelines for municipal AED deployment. Our empirical findings point to four concrete recommendations for Virginia Beach and comparable mid-sized cities. First, building access hours are the single highest-leverage operational lever available to planners. Plans must account for them, and where possible AEDs should go in 24/7 outdoor street cabinets: on held-out arrests, 25 cabinets delivered as much expected benefit as 50 indoor units subject to closing schedules, and 100 cabinets outperformed 200 indoor units (Table 3: 38.2 versus 28.4 potential survivors). A cabinet is worth buying whenever it costs less than about twice an indoor installation. Second, siting optimization should evaluate walking networks with continuous survival-decay scoring rather than an arbitrary 100-meter cutoff. An AED's value halves within roughly 120 meters of walking, but walks of 100 to 250 meters still often reach patients ahead of arriving ambulances, and the classic cutoff ignores them entirely. Third, municipal programs should avoid chasing historically slow EMS response zones. Dispatch latency in Virginia Beach is dominated by operational variance (such as crew availability and traffic) rather than persistent geographic deficits, leaving historical response times unable to guide future siting. Planners need only the city's overall response-time distribution, which most EMS agencies already publish. Finally, cities should adopt an equity floor across social vulnerability quartiles: mandating 85–90% parity cost at most 2.6% of total benefit in-sample, and had no detectable cost on held-out arrests. Because commercial host buildings are scarce in low-density suburban tracts, reaching full parity will require pairing indoor units with street-side cabinets, dispatched volunteer responders, or drone delivery.

Estimated clinical impact. Translating potential survival gains into projected lives saved requires adjusting for clinical arrest presentations in the field. CARES registry data indicate that roughly 10.8% of non-traumatic OHCAs nationwide are bystander-witnessed with a shockable rhythm [2]. Applying this benchmark to the out-of-sample results, a deployment of 50 well-sited indoor AEDs corresponds to approximately 2.0 additional survivors across the 30-month study period (Table 2), and housing 50 units in 24/7 outdoor cabinets raises this to approximately 2.8 (Table 3). These figures represent practical upper bounds: even when a device is accessible, by-

standers may hesitate, fail to recognize cardiac arrest, or lack retrieval assistance. Moreover, 71% of cardiac arrests occur in private residences [2], beyond the reach of any fixed public AED. Public access defibrillation cannot replace baseline EMS capacity, but it is a valuable complement during the critical early minutes of public shockable arrests.

Limitations and future directions. Several methodological constraints qualify our findings. First, the open dispatch data record 911 calls rather than confirmed cardiac arrests, omitting initial cardiac rhythms, witness status, and neurological outcomes; we therefore evaluated survival using logistic survival models from the published literature [4, 5]. Valenzuela et al. reported that their curve overestimates survival at long response times [6], which is where most of the modeled gain lies, so absolute gains may be overstated; the comparison between methods is affected less, since all are scored with the same curve. Second, incident addresses were rounded to the block level for privacy, and several frequent addresses represent healthcare facilities; sensitivity analyses excluding the top 20 addresses confirmed that our model still beats the classic one. Third, opening hours for 720 of the 929 candidate sites were assigned by facility category, so the size of the opening-hours effect depends on these assumed schedules; verifying the hours of the most frequently chosen sites would be the most valuable next data-collection step. Fourth, 374 missing response times were imputed via road distance to the nearest station ($r = 0.21$); excluding these records did not change the ranking of methods. Fifth, our scoring assumes a single willing bystander who knows where the nearest open AED is, and starts CPR only when the AED or EMS arrives. In reality, bystander AED use remains rare [1], scaling realized benefit down across all methods. Sixth, we plan from scratch and ignore any AEDs the city already has. The 51 arrests more than 150 m from any walkable path score zero under every method; 41 of them fall in Q_1, Q_2 and Q_4 (5, 23 and 13 arrests) and the other 10 have no SVI score, slightly lowering those quartiles' maximum benefit C_q, but they do not favour any method. Finally, every method is scored with the same survival model that ours optimizes; we partly guard against this by also reporting simple 100 m coverage, on which the classic method leads at small budgets and the two are tied from $p = 100$.

Future investigations should link municipal spatial data directly to clinical outcomes through the CARES registry, model time-varying bystander availability using mobile mobility data, and extend the integer-programming framework to co-optimize mixed portfolios of fixed indoor units, outdoor street cabinets, and autonomous drone delivery.

7 Prototype: AED Placement Explorer

To translate our optimization framework into an operational decision tool, we developed an interactive web map application (*AED Placement Explorer*, a standalone HTML/JavaScript interface in the project repository). Planners can adjust a budget slider to add AEDs sequentially in greedy order ($p = 1$ to 200) and toggle dynamically between indoor sites with empirical opening hours, 24/7 outdoor cabinets, and the classic 100-meter MCLP. The map displays all 2,706 cardiac arrests colored by EMS response time alongside selected host facilities, annotating each site with its operating schedule and marginal survival contribution. A side panel reports cumulative expected survival gain, percentage improvement over the classic covering baseline, and benefit distribution across social vulnerability quartiles. The complete analytical pipeline—including data preprocessing, walking network routing, mathematical optimization, cross-validation, and figure generation—is open source and fully reproducible from public records.

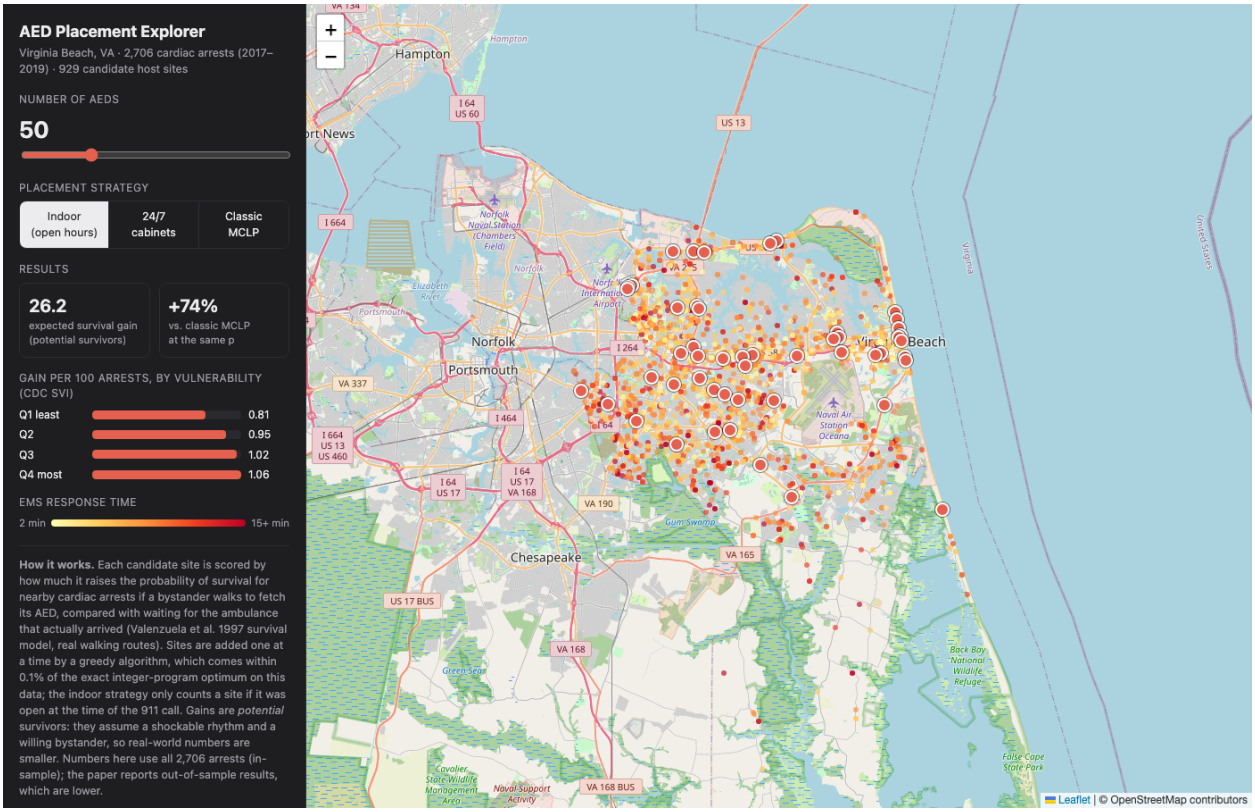


Figure 8: The AED Placement Explorer with 50 indoor AEDs. Numbers in the tool use all arrests (in-sample), so they are higher than the out-of-sample results in Section 5.

8 Conclusion

By coupling municipal open records with clinical resuscitation curves, we demonstrated that an expected-survival, opening-hours-aware optimization model delivers 42–67% greater out-of-sample survival gains than traditional maximal covering formulations, while planning without operating schedules overestimates realized clinical efficacy by approximately 70%. Operational evaluations reveal that a single 24/7 outdoor cabinet provides the clinical value of roughly two indoor AEDs, and that effective device utility declines by half within roughly 123 m of brisk walking. We also found that two intuitive ideas do not hold in Virginia Beach: that AEDs should follow historically slow EMS, and that the most vulnerable neighborhoods are the least served. The broader lesson is that *when* an AED can be reached matters as much as *where* it is, and that some gaps in access cannot be closed by placement alone.

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